

Dhanuja Wanniarachchi

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CONTACT INFORMATION

Name: Wanniarachchige Dhanuja Tharith Wanniarachchi

Title: Dr.

Office mailing address: 47/B, Weralugampola, Papiliyawala, Sri Lanka 11741.

Email: dhanujaw.2020@phdcs.smu.edu.sg

Contact number: +65 98109829

Current position: Recent Ph.D. Graduate (Advised by: Prof. Archan Misra)

EMPLOYMENT HISTORY

PhD Research Intern

Sep 2024-Dec 2024

Institute of High Performance Computing, A*STAR, Singapore

Advisor: Dr. Vigneshwaran Subbaraju

Research Engineer

Nov 2019-Dec 2020

Living Analytics Research Centre (LARC), Singapore Management University

Advisor: Prof. Archan Misra

Verification IP Application Engineer specialised in SPI,I2C,I3C protocols

Feb 2018-June 2019

Synopsys, Sri Lanka

R&D Intern

july 2016 - Jan 2017

MillenniumIT Software (Private) Limited

TEACHING EXPERIENCE

Senior Teaching Assistant

Aug 2025 -Dec 2025

Course Name: Data Security and Privacy

School of Computing & Information Systems Singapore Management University

Senior Teaching Assistant

Jan 2025-Apr 2025

Course Name: Mobile & Distributed Systems

School of Computing & Information Systems Singapore Management University

Senior Teaching Assistant

Jan 2025-Apr 2025

Course Name: Heuristic Search and Optimisation

School of Computing & Information Systems Singapore Management University

Teaching Assistant

Jan 2024-Apr 2024

Course Name: IoT: Tech Appln

School of Computing Information Systems, Singapore Management University

ACADEMIC QUALIFICA- TIONS

Teaching Assistant

Aug 2023-Nov 2023

Course Name: Interconnection of CPS

School of Computing & Information Systems, Singapore Management University

Ph.D. in Computer Science

2021 Jan-2025 Dec

Singapore Management University

Advisor: Prof. Archan Misra

GPA: 3.63

**RESEARCH
INTERESTS**

Collaborative AI; multimodal sensing and sense-making; mobile and distributed systems; deep learning; and reinforcement learning-based real-world applications.

**JOURNAL
CONFERENCE
PUBLICATIONS**

Key: [C] Conference, [J] Journal

[C1] Kasthuri Jayarajah, **Dhanuja Wanniarachchige**, Tarek Abdelzaher and Archan Misra, 2022. ComAI: Enabling Lightweight, Collaborative Intelligence by Retrofitting Vision DNNs. IEEE International Conference on Computer Communications , INFOCOM 2022 [Full Paper]

[J1] **Dhanuja Wanniarachchige** and Archan Misra. 2025. "MIMIC: AI and AR-enhanced Multi-Modal, Immersive, Relative Instruction Comprehension." Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, IMWUT [Journal Paper]

[C2] **Dhanuja Wanniarachchige**,Kasthuri Jayarajah,Dulaj Weerakoon,Tarek Abdelzaher and Archan Misra, 2026. SteerCam: Multi-Camera Edge Perception via Dynamic Joint Steering Collaboration" To appear in IEEE International Conference on Computer Communications , INFOCOM 2026 [Full Paper]

[C3] **Dhanuja Wanniarachchige**,Kasthuri Jayarajah,Tarek Abdelzaher and Archan Misra, 2026. FusionBridge: Enhancing Multi-View Multi-Modal Sensing and Perception for Edge Intelligence" To appear in ACM Conference on Embedded Networked Sensor Systems (SenSys) 2026 [Full Paper]

[J2] Amit Sharma, Vithurson Shubasharan, Manoj Gulati, **Dhanuja Wanniarachchige** and Archan Misra. 2025. "CollabCam: Collaborative Inference and Mixed-Resolution Imaging for Energy-Efficient Pervasive Vision." ACM Transactions on Internet of Things, TIoT [Journal Paper]

[C4] **Dhanuja Wanniarachchige**,Mayooran Thavendra,Kasthuri Jayarajah,Archan Misra. "MultiSense-RL-Arena: A Unified Reinforcement Learning Environment for Multi-View and Multimodal Perception" KDD Conference (under review) [KDD 2026 Datasets and Benchmarks Track]

[C5] **Dhanuja Wanniarachchige**, Vigneshwaran Subbaraju, and Archan Misra. "ViSoR-BC: An Explainability-by-Design Approach to Engagement Estimation in Videos via Social Signals." (under review) in IEEE International Conference on Multimedia and Expo (ICME) 2026.

**HONORS &
AWARDS**

2nd place in Kaggle competetion for Product Detection

Hosted by Shopee, Singapore

2nd place out of 646 teams

Dean's list in Semesters 6,7 & 8 of B.Sc. degree

Semester6 GPA 3.81

Semester7 GPA 3.82

Semester8 GPA 4.15

REFERENCES

Lee Kong Chian Prof. Archan Misra

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Assistant Professor, New Jersey Institute of Technology
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